On-Device Continual Personalization for RSS Readers: A TinyML Approach to Privacy-Preserving Recommendation

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Abstract

This draft proposes an on-device recommender framework for RSS reading that performs continual personalization using only local user behavior. The objective is to mitigate model collapse and privacy risks by keeping interaction logs, feature extraction, adaptation, and model updates entirely on-device. The work targets Android first and is designed to remain compatible with future TinyML transfer constraints for STM32-class hardware. The current implementation path prioritizes a deployable local baseline with strict efficiency constraints on latency, memory footprint, storage growth, and battery cost.

Contents

1 Introduction

1.1 Problem Statement

TODO: Define the practical and scientific problem in RSS recommendation under privacy and resource constraints.

1.2 Motivation

TODO: Explain why local-only continual learning is needed and where cloud-first recommenders fail for this context.

1.3 Contributions

TODO: Add concrete contributions after first stable experiment cycle.

2 Background and State of the Art

2.1 On-Device Recommender Systems

TODO: Summarize key approaches from the DeviceRS survey.

2.2 Continual Learning at the Edge

TODO: Summarize edge continual training assumptions and limits.

2.3 TinyML Online Learning

TODO: Extract STM32-relevant constraints and lessons from TinyOL.

2.4 Privacy-by-Design in Mobile Systems

TODO: Map privacy principles to this architecture.

3 Research Gap and Novelty

3.1 Gaps in Related Work

TODO: Fill from matrix in related-work-matrix.md.

3.2 Proposed Novelty

TODO: Define novelty claims and boundary conditions.

3.3 Deployment-Centric Claims

TODO: Formalize efficiency-first claims with measurable thresholds for inference latency, online update overhead, memory footprint, and battery impact.

4 Methodology

4.1 System Architecture

TODO: Add architecture figure and component breakdown.

4.2 Selected Implementation Approach (Current)

- **Phase A (current):** Lightweight local continual learner with event-driven updates and bounded replay, fully on-device.
- **Phase B:** Drift-aware adaptation policy using short-term vs long-term behavior divergence with update scheduling.
- **Phase C:** TinyML transfer track with compact model/state representation aligned with STM32 constraints.

4.3 Local Interaction Pipeline

TODO: Define events (impression, open, read-session) and retention policy.

4.4 Continual Learning Strategy

TODO: Describe baseline learner, replay, drift logic, and anti-collapse safeguards.

4.5 Efficiency and Footprint Constraints

TODO: Report budgeted constraints and observed values:

- Inference latency per ranked item and per list refresh.
- Local update latency and scheduling frequency.
- Memory and persistent storage footprint growth over time.
- Battery and thermal overhead in realistic usage sessions.

4.6 Privacy Model

TODO: Threat model and guarantees for local-only design.

5 Experimental Protocol

5.1 Research Questions and Hypotheses

TODO: Add RQ1–RQn and corresponding hypotheses.

5.2 Datasets and Real-Usage Collection

TODO: Describe realistic data collection without mock data.

5.3 Baselines and Ablations

TODO: Static baseline vs CL-v1 and ablations.

5.4 Metrics

TODO: Utility, diversity, drift resilience, energy, memory, and latency.

5.5 Primary Evaluation Targets

- Accuracy and ranking quality under non-stationary user behavior.
- Adaptation speed after concept drift events.
- Resource efficiency: latency, memory footprint, storage overhead, energy cost.
- Privacy robustness under local-only constraints and no raw behavior export.

6 Preliminary Implementation Status

- v2.0.1: Added local on-device interaction event pipeline in app code.
- v2.0.1: Added feed-level impression/open logging and reader dwell-depth session logging.
- v2.0.1: Added settings controls for local learning enablement, retention, summary, and reset.

7 Threats to Validity

TODO: Internal, external, construct, and conclusion validity.

8 Ethics and Privacy Considerations

TODO: User consent, transparency, and local data governance.

9 Conclusion and Next Steps

TODO: Summarize expected outcomes and near-term milestones.